

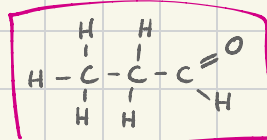
17 - Carbonyl compounds

04/01/25

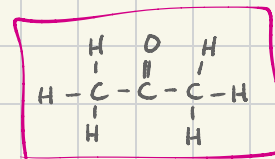
Carbonyl compounds

→ compounds that have at least one $C=O$ bond

e.g. aldehydes & ketones



aldehyde



ketone

Production of aldehydes & ketones

i) aldehydes

* formed via the oxidation of primary alcohols using acidified $K_2Cr_2O_7$ / $KMnO_4$ and distillation

ii) ketones

* formed via the oxidation of secondary alcohols using acidified $K_2Cr_2O_7$ / $KMnO_4$ and distillation

Reactions of aldehydes & ketones

i) reduction

reagents: $NaBH_4$ or $LiAlH_4$

aldehydes → primary alcohols

ketones → secondary alcohols

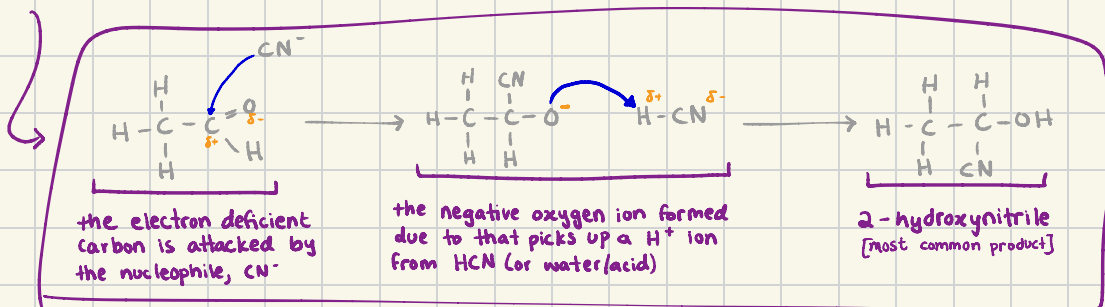
ii) reaction with HCN

reagents: KCN as a catalyst (and heat)

produces hydroxynitriles



nucleophilic addition

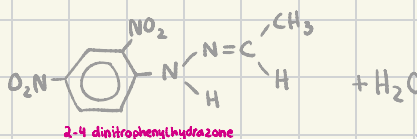
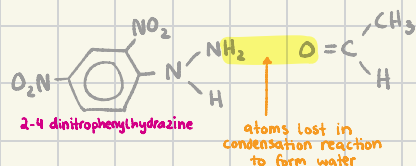


Tests for aldehydes & ketones

• 2,4-dinitrophenylhydrazine (2,4-DNPH)

→ carried out using Brady's reagent (solution of 2,4-DNPH in methanol & sulfuric acid)

→ if aldehyde/ketone is added to Brady's reagent, an orange yellow ppt. is formed



test to determine whether an unknown compound is an aldehyde or ketone

• Tollens' reagent (silver mirror test)

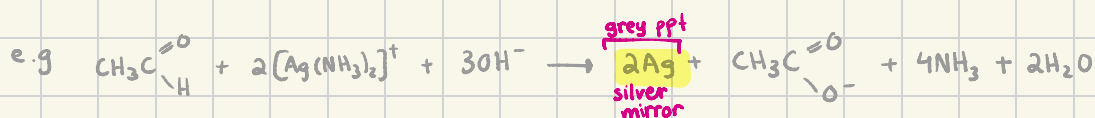
→ positive result (a grey ppt. forming due to silver mirror) if compound is an aldehyde

→ aldehydes reduce the diaminesilver(I) ions to metallic silver & itself is oxidised to a salt of carboxylic acid

→ ketones cannot be oxidised, thus cannot reduce it to metallic silver

making of Tollens' reagent

- ① prepare a solution silver nitrate 
- ② add NaOH solution dropwise until a light brown ppt. forms 
- ③ add conc. ammonia solution until the brown ppt. of Ag₂O just redissolves (silver ions have been complexed into [Ag(NH₃)₂]⁺ ions) 
- ④ add a few drops of suspected aldehyde & put solution in a warm water bath 



• Fehling's reagent

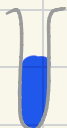
* contains copper (II) ions complexed with tartrate ions in NaOH solution

* complexing the copper (II) ions with tartrate ions prevents precipitation of copper (II) hydroxide

→ positive result (a red ppt. forming) if compound is an aldehyde

→ aldehydes reduce the complexed copper (II) ions into copper (II) hydroxide & itself oxidises to a salt of carboxylic acid

→ ketones cannot be oxidised, thus cannot reduce it to copper (II) hydroxide

Fehling's solution  → put in warm water bath after adding drops of sus aldehyde  → positive result red ppt. 

• Reaction with alkaline I₂ (aq)

→ identifies the presence of a CH₃CO- group in an aldehyde or ketone

① the carbonyl compound is halogenated; the three H atoms in the CH₃ methyl group are replaced by I atoms

② the intermediate is hydrolysed to form a yellow ppt. of tri-iodomethane

→ positive result (a yellow ppt. forms) if compound is ethanal or any methyl ketone

